

solution #2: Advantage functions

⇒ The advantage function measures how much better an action is compared to the policy's average behavior in a given state, turning absolute rewards into meaningful learning signals

⇒ equations: $V(s) = E[R | s]$ = expected reward from state s

value function

↪ if I sample a response from current policy at prompt s, what reward do I expect on average?

expected reward

$Q(s, a) = E[R | s, a]$ = force the model to produce response a from prompt s

action value function

↪ if I force the model what reward do I expect to produce exactly response a for prompt s,

$Q(s, a) = R(s, a)$ what reward do I expect?

↓ Definition (advantage): $A(s, a) = Q(s, a) - V(s)$

↪ how much better (or worse) this action is compared with the average action for this state?

↪ why baseline function = value function gives advantage.

⇒ if: $b(s) = V(s) = E[R | s]$

Then: $R(s, a) - b(s) = R(s, a) - V(s) = A(s, a)$

So: baseline reward = advantage